

HITESH NANDAKUMAR

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EDUCATION

California Polytechnic State University, San Luis Obispo

Expected May 2028

Bachelor of Science in Computer Engineering, Concentration in Embedded Systems | Minor in Ethics, Public Policy, Science, and Technology

GPA: 3.73 | Dean's List, 5 of 6 quarters

TECHNICAL SKILLS

Programming: C/C++, Python, Java, C#, JavaScript, RISC-V (RV32I) Assembly

Hardware & Digital Design: SystemVerilog, RTL Design, RISC-V (RV32I), Timing Analysis, Waveform Debugging, Vivado

Embedded & Electronics: STM32, GPIO, Sensor Interfacing, Circuit Debugging, Oscilloscope, Logic Analyzer, DMM, Signal Generator, LTspice

Systems & Software: POSIX (fork, mmap, shared memory), Git/GitHub, JUnit, UML

ENGINEERING PROJECTS

32-bit RISC-V Pipelined Processor (OTTER MCU)

CPE 233 / CPE 333

SystemVerilog | Computer Architecture | 2-Person Team

- Co-designed and built a 5-stage pipelined RV32I processor with a partner, jointly developing the datapath, control logic, and hazard-handling units
- Designed forwarding and load-use stall logic to resolve data hazards, and branch-flush logic to squash wrong-path instructions on taken branches
- Authored RISC-V assembly test suites and self-checking testbenches; used waveform debugging to isolate and correct functional and timing failures

FSK IR Communication System & Low-Voltage Sensing Circuit

EE 115/145, EE 215/245

Analog Circuit Design | LTspice | Experimental Validation

- Designed, simulated, and physically built an FSK-based IR communication system; tuned a transimpedance-amplifier front end in LTspice to keep two FSK tones cleanly separable, then validated on the bench
- Designed and validated a low-voltage detection circuit with datasheet-driven component selection; quantified performance via voltage-sweep testing (8.02V trip, -6.3mV error) in an IEEE-format technical report

Systems Programming in C: Parallelism & IPC

CSC 357

C | POSIX APIs | Concurrency

- Parallelized image-processing operations (rotation, channel swap) across multiple fork-based processes sharing memory via anonymous mmap, with measured runtime speedup
- Wrote a custom atomic-flag spinlock from scratch to safely coordinate shared state across concurrent processes without a library mutex
- Built a two-process client/server using POSIX named shared memory (shm_open/mmap) for inter-process image processing

LEADERSHIP & EXPERIENCE

Resident Advisor (RA)

August 2025 – Present

California Polytechnic State University, San Luis Obispo

- First point of contact for 50+ undergraduate residents, responding to emergency situations, managing conflict resolution and community initiatives; receiving an award for organizing an outstanding community event

Swim Instructor

November 2023 – September 2024

American Swim Academy, Livermore, CA

- Taught swim lessons to 75+ students ages 3–17 over ~3 years, working 15–25 hrs/week while maintaining a full high school academic and athletic course load
- Identified a child in distress in the deep end when another instructor did not notice and independently performed a water rescue